



MBSE and Requirements Analysis, Key to Successful Systems Engineering

Macaulay Osaisai

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- ❖ Definitions
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- ❖ The case for MBSE implementation and requirements analysis
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Introduction

❖ Why MBSE Implementation and Requirement Analysis

- Every product or service (system) needs a rationale for acquisition, development, modification, expected operational environment, and usage.
- MBSE and requirements analysis together provide the tools and workflow to examine, evaluate, and translate customer needs statements into a requirements baseline and model framework for system definition, design, and development.

❖ Requirements Analysis

- Requirements analysis is the primary activity in the requirements life cycle phase.
 - ✓ Requirements analysis is repeated for system, preliminary, and detailed design phases.
- Requirements analysis, a.k.a. requirements engineering, is the process of **identifying and validating user needs** and **transforming those needs into system requirements**.
- **Requirements validation** is the critical communication activity in which customer, user, and developer stakeholders work iteratively to **ensure a common requirements understanding and interpretation**, from which an acceptable solution can be developed.

Definitions

❖ Definition of a Requirement

- A requirement¹ is a specification of a need or want. Sets of requirements are used to capture the information needed to design, build, and test a process, service, product or system. Each individual requirement is typically structured to be distinct, relevant, testable, traceable and unitary; meaning that it addresses one thing and only one thing. Collectively, sets of requirements are typically designed to be consistent and cohesive. In some cases, requirements are mapped to things such as business goals, business units and anything else that needs to be traceable. ¹<https://simplicable.com/new/requirement-definition>
- A requirement is a statement that identifies a product or process operational, functional, or design characteristic or constraint, which is unambiguous, testable or measurable, and necessary for product or process acceptability (ISO 2007).
- “A statement that identifies a product or process operational, logical, or design characteristic or constraint, which is unambiguous, testable or measurable, and necessary for product or process acceptability (by consumers or internal quality assurance guidelines)” (IEEE-STD-1220, 2005).

Objectives

❖ Requirements Objectives

- Establish the need for a product or service
- Define the system operational environment, usage, and scenarios for operations
- Define functional behavior, performance, inputs/outputs (interfaces), and data
- Define standards and constraints for reliability, maintainability, security, logistics, deployment, usability, transportation, etc.

The case for MBSE implementation and requirements analysis

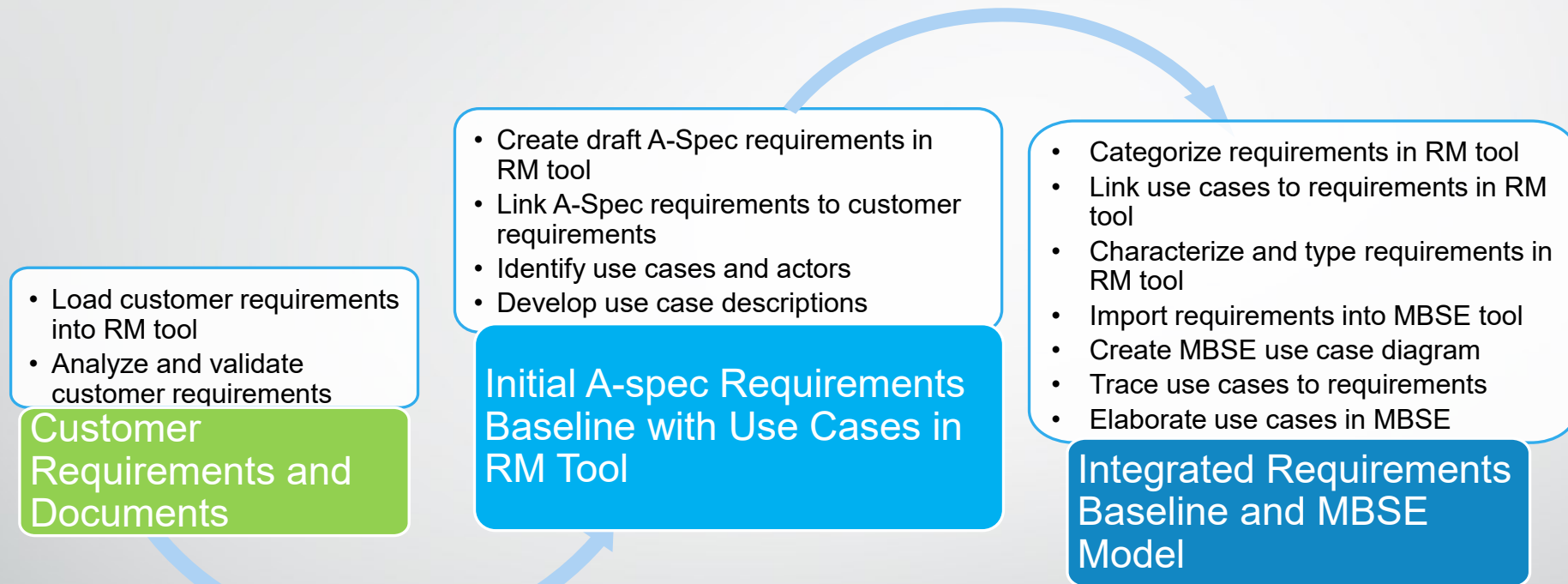
- ❖ **Systems development demands disciplined requirements analysis**
- ❖ **Requirements must be traceable from customer needs statements to**
 - System functionality (via use cases)
 - System, subsystem, and configuration Item design requirements
 - System architecture and design models, including interfaces
- ❖ **MBSE workflows combine modeling tools with Requirements Analysis tools and processes to create models with all required traceability**
- ❖ **MBSE / Requirements Analysis is tailorable to operate at the desired level based on program need (cost, schedule, long-term sustainability needs)**
 - Top level system requirements (A-Specs) analysis –SRR baseline
 - ✓ Mapping to use cases and top-level architecture
 - ✓ Allocating to subsystems
 - ✓ Tracing to customer needs/requirements
 - Subsystem level analysis
 - ✓ B-Spec development and allocating to CI-level requirements and design
 - ✓ Any desired lower level (e.g., firmware, CCA design, FPGA design)

Requirement analysis using MBSE

- ❖ Load customer requirements into a Requirement Management (RM) tool/environment
- ❖ Analyze customer requirements to understand the operational context, scenario and usage
- ❖ Create draft A-Spec requirements in RM tool/environment
- ❖ Link draft A-Spec requirement to customer requirements in RM tool/environment
- ❖ Identify stakeholders, actors, and use cases
- ❖ Develop a description of each use case that address the operational context, scenario and usage of all aspects of the system based on the requirements and other pertinent documents
- ❖ Create attributes to characterize A-Spec requirements. Typical attributes: Requirement Category, Use Case, and Requirement Type
- ❖ Categorize every A-Spec requirement as either TRUE (is a requirement) or FALSE (not a requirement)
- ❖ Trace each use case to a specific set of requirements that is consistent with the use case description and scenario
- ❖ Characterize and type all A-Spec requirement that are categorized “TRUE”. Typical requirement types are: Functional, Performance, Interface, Physical, Standard Constraints, RMA Business

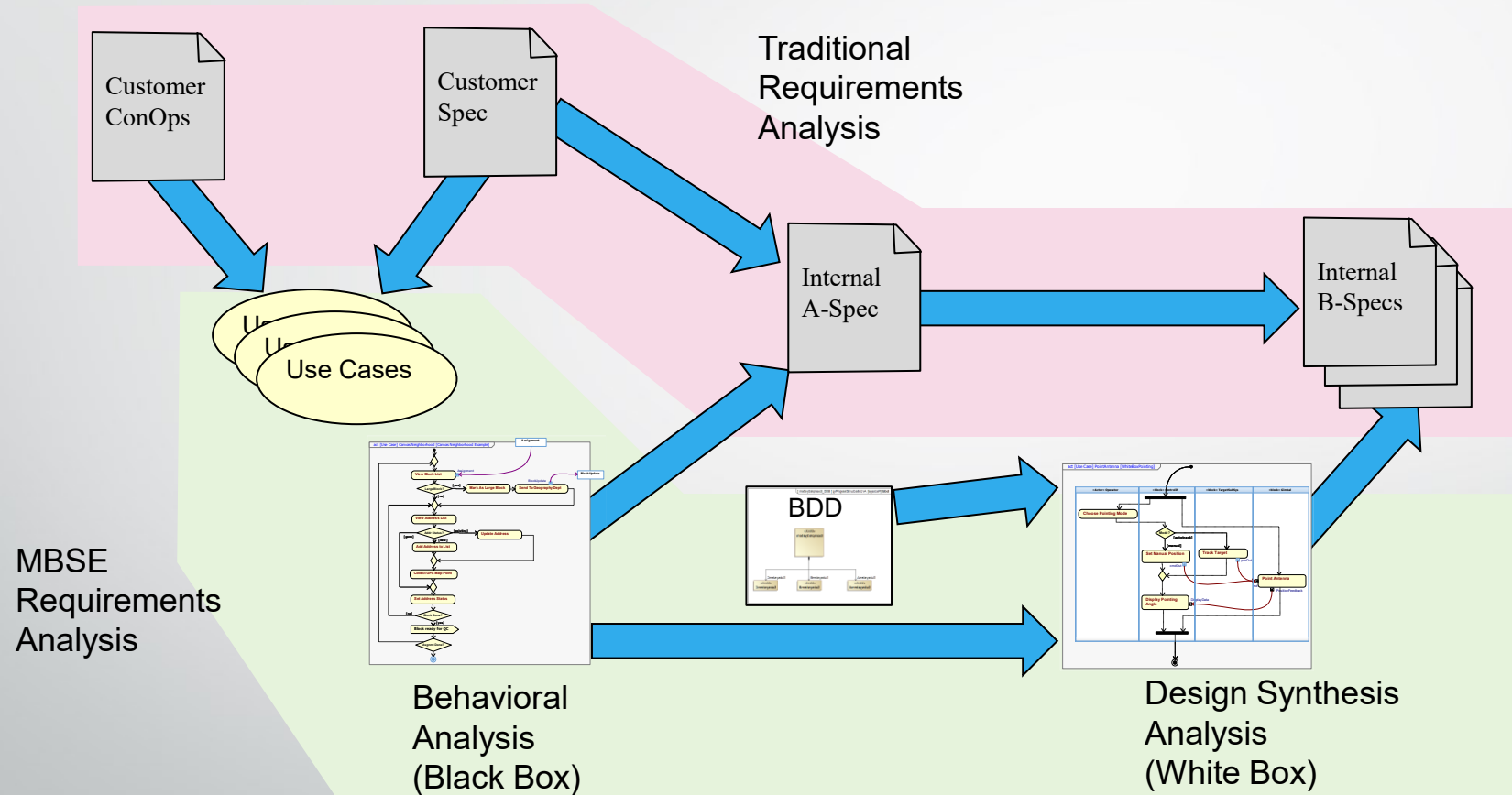
Use Case Analysis Ensures Robust Requirements Baseline Results

Requirements analysis workflow – MBSE implementation



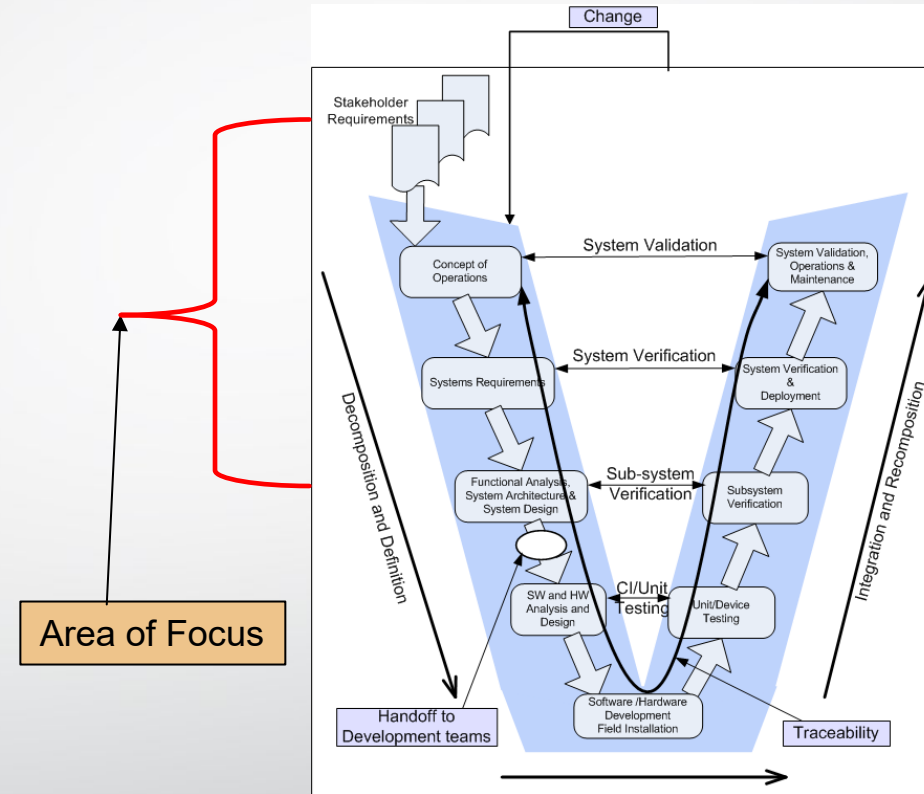
Use Case Analysis Enhances Requirements Analysis and Improves System Engineering Process

MBSE approach and traditional requirements analysis



MBSE in context of the system VEE Model

- ❖ The system development life-cycle (VEE Model)
- ❖ MBSE approach adheres to this life-cycle
- ❖ MBSE is not a new way of doing system engineering
- ❖ MBSE augments system engineering



MBSE Augments Traditional Requirements Analysis

Attributes in RM tool/environment -1

❖ **Attribute**

- Tag that characterizes RM tool/environment entries

❖ **Requirement Attribute - Boolean**

- Categorize each RM tool/environment entry
 - ✓ TRUE – The RM tool/environment entry is a requirement
 - ✓ FALSE – The RM tool/environment is a non requirement, such as headers and other information

❖ **Use Case Attribute**

- Traces use cases to all functional requirements and selected requirements of other types
- Provides a bookkeeping capability since all requirements are traced to a specific use case
- No more than one use case should be traced to a functional requirement; therefore, any occurrence of multiple use cases tracing to a single functional requirement is an indication of the existence of a compound requirement that requires further decomposition.

Attributes in RM tool/environment -2

❖ Requirement Type Attribute

- Enables characterization of each system requirement categorized to be “TRUE”. In the literature, it is alluded that a single functional requirement could have up to 15 associated requirement types. Consequently, this attribute provides the means to type and characterize every system requirement at the onset. The process enforces discipline and rigor; every requirement in the system requirements set must be examined so full coverage is attained.
- Various requirement types.

	Business Requirement
	Design Constraint
	Extended Requirement
	Functional Requirement
	Interface Requirement
	Performance Requirement
	Physical Requirement
	Usability Requirement

Source: MagicDraw (No Magic documentation)

Use case analysis of system behavior

❖ Use cases describe system behaviors in response to actor (stakeholder) stimulus

- Primary stakeholders provide inputs to the system
- System responds
- Different sequences (scenarios) of behavior can take place
- Use case collects related scenarios

❖ A system may have a few to many use cases

❖ Normally text

❖ Use case description and scenario affirm trace to requirements

❖ Use case diagram identifies user stakeholders and their goals

❖ Supports system threads

• Typical UC Description Template

```
Use Case:      <Name>
Actor(s):      <>
Preconditions: <>
Post-conditions: <>
Dependencies:  <>
Includes: <>
Description:   <>
Scenario:
<>
Alternate:
<>
```

Use Case Analysis Enhances Requirements Analysis

Use case description

Use Case:	TaskSystem
Actor(s):	BattlefieldCommander, Operator
Preconditions:	BattlefieldCommander has gained authorized access to the system.
Postconditions:	The system has accepted and stored the tasking order and notified the operator that an order is awaiting review.
Dependencies:	NA
Description:	The BattlefieldCommander access the system and sends a 1) information on collateral emitters, and 2) a task order to the system operator. Collateral information provides information on the expected location of blue-force emitters and known locations of infrastructure emitters, as well as signal characteristics for each. The task order indicates RF characteristics to monitor. The task order includes the following information: 1. Communications modes 2. Frequency ranges 3. Geographic area 4. Time and duration of order The system verifies the syntax of the Task order and either rejects it as invalid or acknowledges it. The system accepts a valid TaskOrder, stores it for review of the operator, and notifies the Operator an order is waiting.
Scenario:	1. Enter task order 2. Enter collateral information 3. Validate task order 4. Store task order 5. Create notification for operator
Alternate:	1. Notify commander if task order syntax is wrong 2. Notify commander if task order logic is wrong

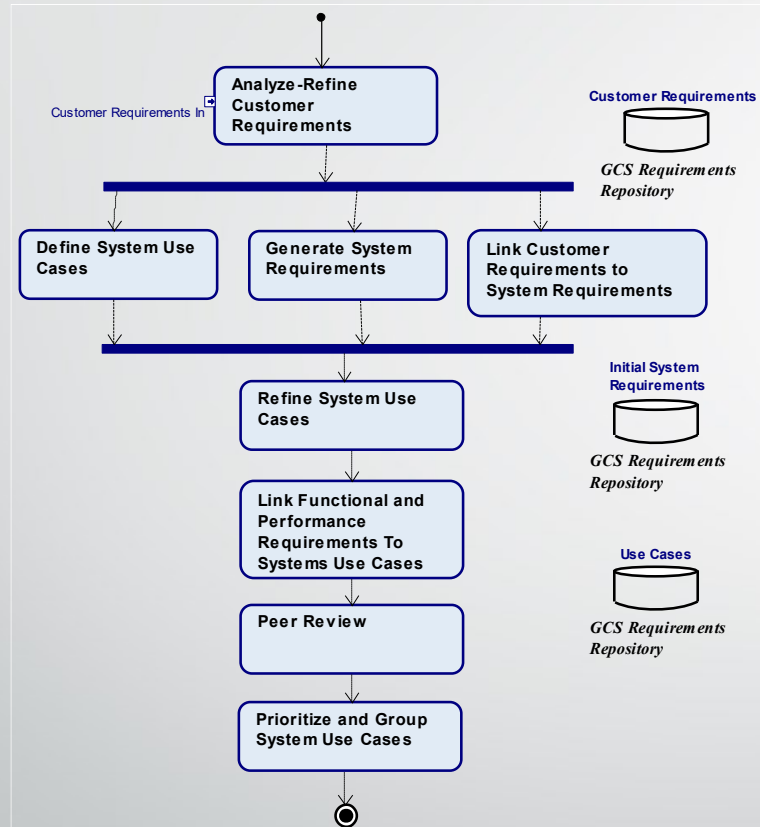
Requirements analysis in RM tool/environment

ID	Requirement Text	Requirement	_ReqType	_Use Case
H_CSE-5	3. The system shall determine the location of a ground-based target transmitter with an accuracy of less than or equal to 0.5 KM	True	Functional Performance	ProsecuteSignals
H_CSE-6	4. The system shall classify signal emitters against a known list at a rate of not less than 100 per second. The known emitter list is provided by the BCD Agency.	True	Functional Performance Guidance Interface	ReportOnMission
H_CSE-7	5. The system shall provide an Emitter Classification Accuracy of less than 2 errors per 1000 classification operations, averaged over a sample of 10,000 operations.	True	Functional Performance	ProsecuteSignals
H_CSE-8	6. The system shall process incoming messages of type XXX at a rate of four (4) per second, sustained for a period of not less than 10 seconds within a 60 second interval	True	Functional Performance	ProsecuteSignals
H_CSE-9	7. The system shall fuse emitter data with collateral information at a minimum rate of 100 signal intercepts per second, when in an automated mode.	True	Functional Performance Constraints	ProsecuteSignals
H_CSE-10	8. The system shall provide automatic and manual modes of data fusion operations	True	Functional	ProsecuteSignals
H_CSE-11	9. The system shall provide the capability for an operator to assume manual control of the fusion of emitter data.	True	Functional Interface	FuseManually
H_CSE-12	10. The system shall perform Query Transactions at a minimum Rate of 10 per second.	True	Functional Performance	QueryForIntel
H_CSE-13	11. The system shall provide access authentication through a user login process.	True	Functional	ConfigureSystem
H_CSE-14	12. The system shall provide management capabilities: User authentication, performance and health monitoring	True	Functional Guidance	ConfigureSystem
H_CSE-15	13. Upon command, the system shall query the BCD Agency for mission-related Intel data.	True	Functional	QueryForIntel
H_CSE-16	14. Upon command, the system shall provide Intel update reports to the BCD Agency	True	Functional	UpdateIntel
H_CSE-17	15. The system shall respond to Battlefield commander queries for mission reports.	True	Functional	ReportOnMission
H_CSE-18	16. The system shall respond to Collateral Consumer queries for mission reports.	True	Functional	ReportOnMission
H_CSE-19	17. Mission reports shall contain the following data: frequency, bandwidth, modulation, data rate, protocol and information content of intercepted signal	True	Functional	ReportOnMission
H_CSE-20	18. The system shall accept tasking messages from the Battlefield Commander.	True	Functional	TaskSystem
H_CSE-21	19. The operator shall load Battlefield Commander tasking data into the system.	True	Functional	ConfigureSystem
H_CSE-22	20. The operator shall configure the system with communication modes, frequency ranges, and geographical area of interest.	True	Functional	ConfigureSystem
H_CSE-23	21. The system shall be operable by personnel with MOSnnnn, as defined in the Military Occupational Specialty Document, Version X, and dated YYYYMMDD.	False		
H_CSE-24	22. The system shall operate in a ground-deployed in a stand-alone or in concert with existing facilities	False		
H_CSE-25	23. The system shall be setup using only commercial tools.	False		
H_CSE-26	24. The system shall operate from -55 ^o to 125 ^o Celsius	False		
H_CSE-27	25. The system shall be in LAW MIL-STD-810G method 520.3 (Temperature, Humidity, Altitude)	False		

Requirements analysis dump from RM tool/environment

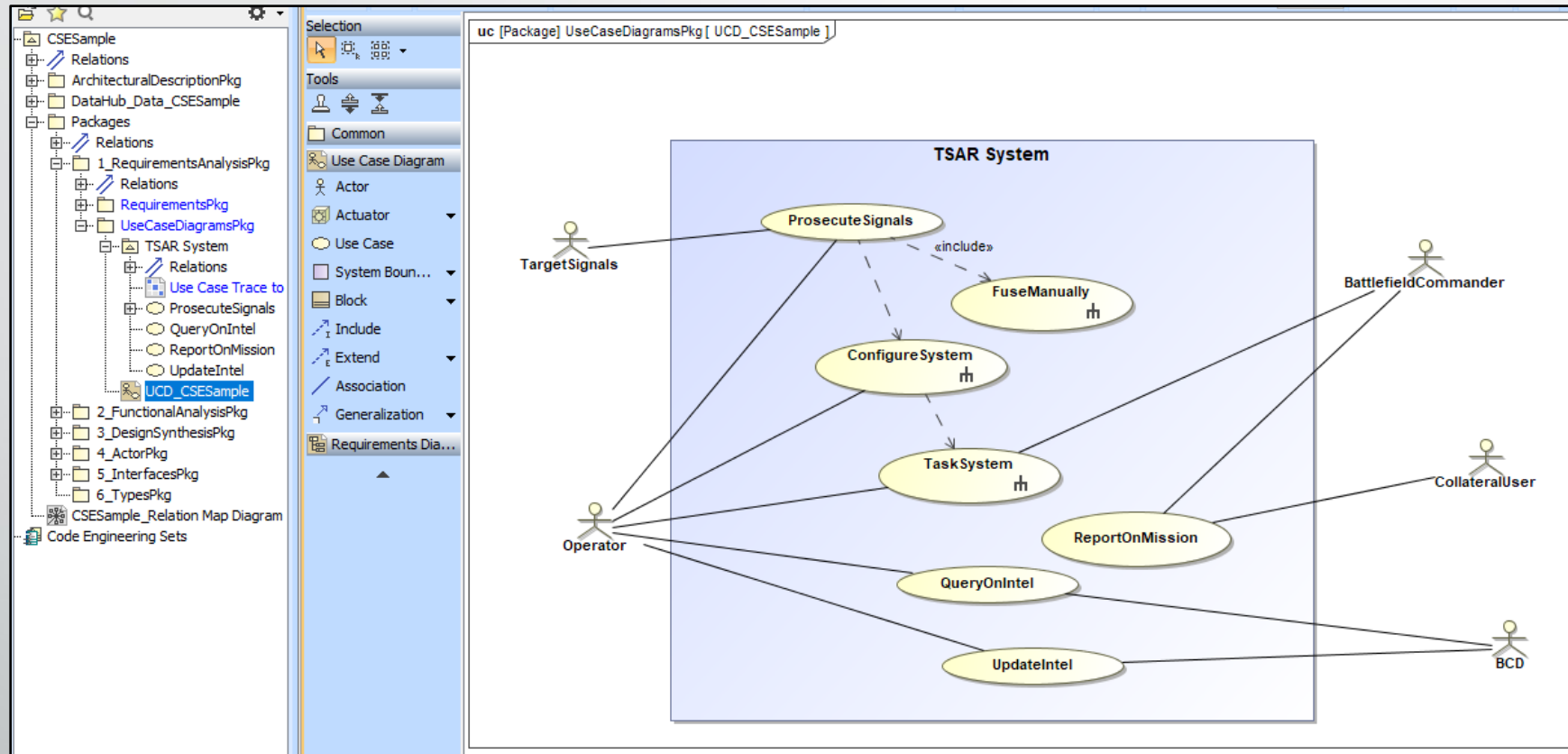
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H_CSE-16	14.Upon command, the system shall provide Intel update reports to the BCD Agency	True	Functional	UpdateIntel
H_CSE-17	15.The system shall respond to Battlefield commander queries for mission reports.	True	Functional	ReportOnMission
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MBSE implementation requirements analysis workflow



- Requirements analysis per standard practice
 - RM Tool/Environment
- PLUS: system use cases per MBSE process
 - Identify use cases
 - Link to functional requirements
 - Identify Actors
 - Define use case scenarios
 - Sunny day
 - Non-nominal / alternate
 - Create use case diagram in SysML tool
 - Populate use cases in SysML tool

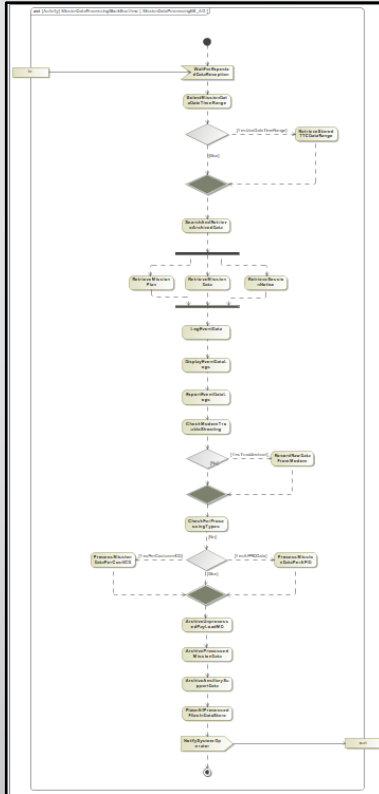
Use case diagram



Requirements table

Criteria		
Scope (optional): 1 A-Specification {} ... Filter: ▼		
#	Name	△ Text
1	[R] H_CSE-1	
2	[R] H_CSE-2 Top Level	
3	[R] H_CSE-3	1. The system shall collect and prosecute ABC satellite downlink signals.
4	[R] H_CSE-4	2. The system shall collect and prosecute signals from the VHF spectrum.
5	[R] H_CSE-5	3. The system shall determine the location of a ground-based target transmitter with an accuracy of <u>less than or equal to</u> 0.5 KM
6	[R] H_CSE-6	4. The system shall classify signal emitters against a known list at a rate of <u>not less than</u> 100 per second. The known emitter list is provided by the BCD Agency.
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15	[R] H_CSE-15	13. Upon command, the system shall query the BCD Agency for mission-related Intel data.

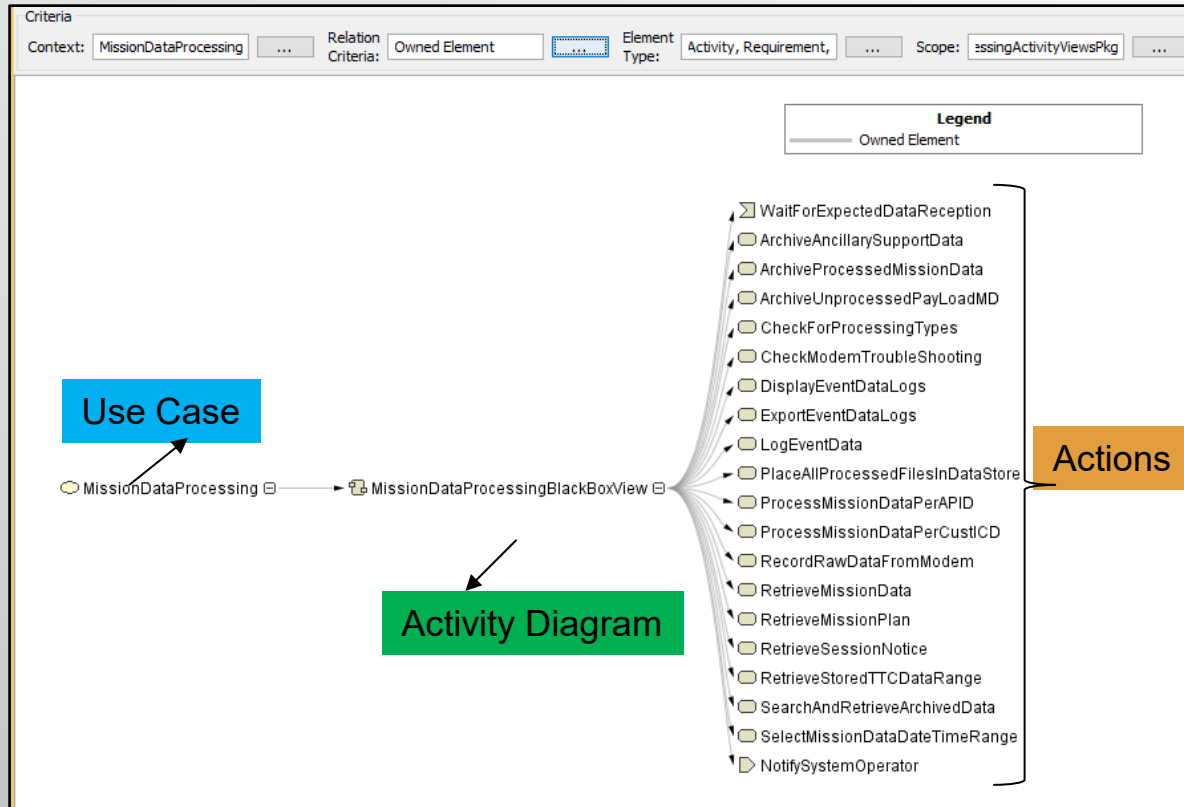
Functional behavior analysis – Activity diagram



- ❖ Activity diagram - Elaboration of a use case
- ❖ Activity diagram exposes requirement gaps and identifies new requirements
- ❖ Activity diagram exposes and resolves compound requirements
- ❖ Activity diagram exposes and resolve duplicate requirements
- ❖ Sequence and state diagrams could also be used to elaborate a use case

MBSE Use Case Elaboration Enhances Requirements Analysis

Relation map diagram



- ❖ Relation
 - Use case
 - ✓ Activity diagram
 - Actions
- ❖ Use case owns the activity diagram
- ❖ Activity diagram owns the actions
- ❖ Underlying relation is standard in MBSE tools

Tool Specific Features Enhance Requirements Analysis

Actions satisfy requirements relation matrix

Criteria

Row Element Type: EventAction, CallBehaviorAction, SendSignalAction ... Column Element Type: Requirement, AbstractRequirement ...

Row Scope: MissionDataProcessing {key} ... Column Scope: 1 Ground Segment {key} ...

Dependency Criteria: Satisfy, Satisfy (Implied) ... Direction: Row to column Show Elements: With relations

Legend

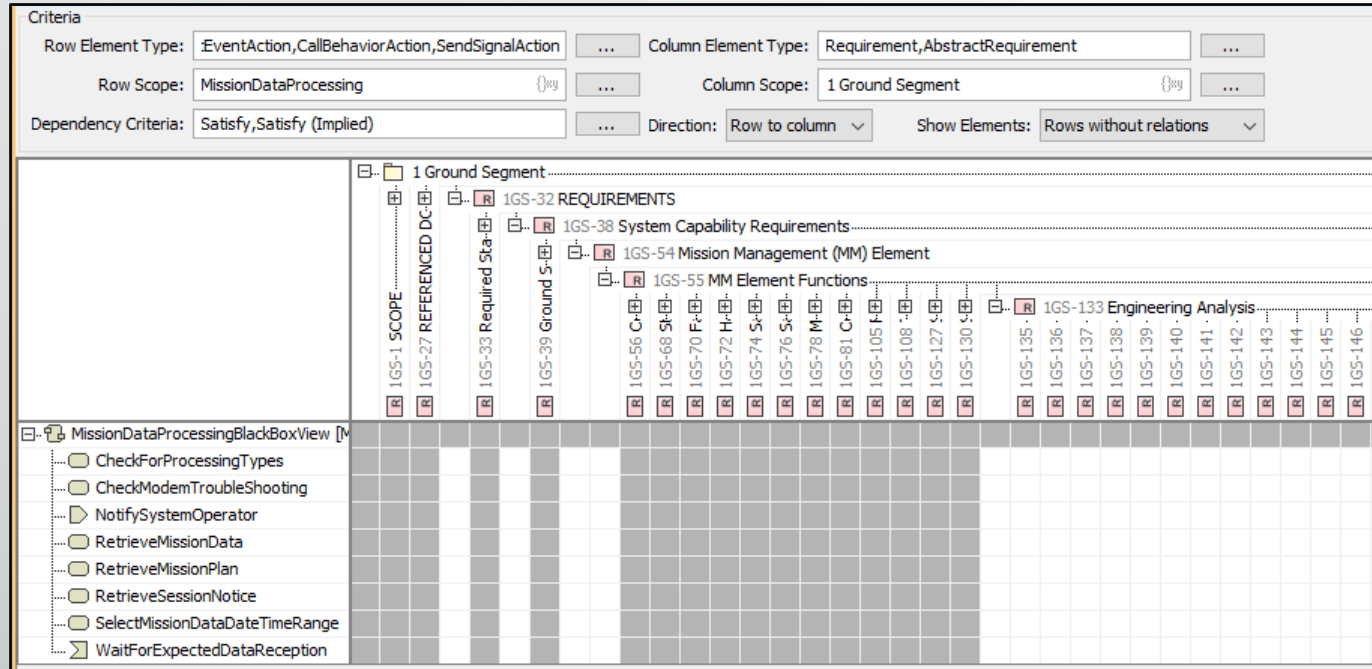
↗ Satisfy

	1G5-38 System Capability R	1G5-55 MM Element Fun	1G5-133 Engineering	1G5-151	1G5-316 MM Event L	1G5-317	1G5-318	1G5-319	1G5-320 Raw Data F	1G5-321	1G5-380 MM and MD	1G5-381	1G5-418 Mission Data Pr	1G5-419	1G5-420	1G5-421	1G5-422	1G5-423	1G5-424
MissionDataProcessingBlackBoxView [M]	1		1		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
ArchiveAncillarySupportData	1											1							↗
ArchiveProcessedMissionData	1											1							↗
ArchiveUnprocessedPayloadMD	1											1							↗
DisplayEventDataLogs	1	1			1														↗
ExportEventDataLogs	1	1			1														↗
LogEventData	1	1			1														↗
PlaceAllProcessedFilesInDataStore	1											1							↗
ProcessMissionDataPerAPID	1	1									1								↗
ProcessMissionDataPerCustICD	1											1							↗
RecordRawDataFromModem	1	1							1										↗
RetrieveStoredTTCDDataRange	1											1							↗
SearchAndRetrieveArchivedData	1	1	1																↗

- ❖ Actions of an activity diagram of a use case satisfy specific requirements
- ❖ An atomic action must satisfy only 1 functional requirement, but may also satisfy other non-functional requirements

Satisfy Relation enhances Requirements Analysis

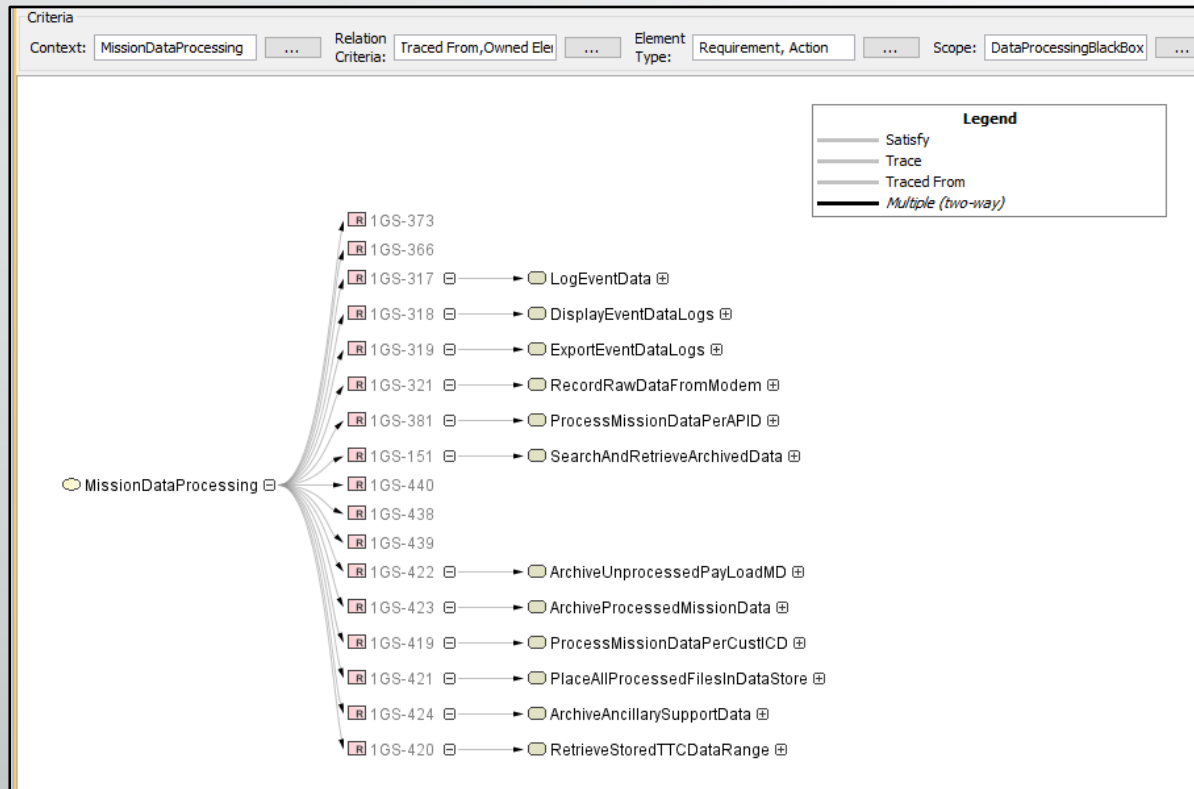
Actions with no requirements to satisfy



- ❖ Orphan actions are an indication of missing requirement
- ❖ Requirements are easily identified early in the design cycle
- ❖ Less cost to update/change requirements

Facilitate Communication and Provide Rigorous Requirements Analysis
Requirement Gaps are Represented Visually in Satisfy Relation Matrix

Use case relation map diagram



- ❖ Use case traced to requirements
- ❖ Requirements satisfied by actions
- ❖ Traceability of requirement to model element is maintained

Visualization Features Improve Requirements Analysis

Lessons learned and future work

❖ Lessons Learned

- Don't be afraid of the tools – use them
- Quickly build your MBSE and RM tool/environment and train the team
- Start using the MBSE SysML tools early
- Use proven requirement management principles

❖ Future Work

- Develop a Model Based Conceptual Design approach for generation of customer requirements during Enterprise Architecture development
 - ✓ INCOSE Model Based Conceptual Design Working Group

Conclusion and remarks

❖ MBSE implementation

- Augments traditional requirements analysis with discipline and rigor
- Does not replace traditional requirements analysis, it complements
- Cultivates and foster effective communication with customer during the requirements analysis phase
- Provides early discovery of requirements gaps, preventing escapes
- Enables early detection of ambiguous and compound requirements
- Results in a more complete and robust requirements baseline



Questions?